

Requirements Categories and Subcategories

		Business Requirements	Technical Specifications
Functional (Doing)		01 Processes / Procedures (Manual)	51 Modules / Programs (Automated)
Informational (Knowing)		11 Knowledge 12 Information 13 Data Descriptions 14 Help facilities 15 Data Persistence	61 Data structures 62 Data bases 63 Access Paths 64 User Views 65 Data Properties
Behavioral (Acting)	Quantitative (Objective)	21 Urgency 22 Frequency 23 Volumes 24 Accuracy	71 Size 72 Precision 73 Response Time 74 Update Time
	Qualitative (Subjective)	31 Usability 32 Trainability 33 Flexibility	81 Maintainability 82 Reliability 83 Portability 84 Scalability
Environmental (Being)		41 Distribution (Organization) 42 Location 43 Security 44 Legal 45 Cultural	91 Interfaces (Interaction) 92 Standards 93 Infrastructure 94 Methodology 95 Technology

Requirements Taxonomy

Business Requirements	Definitions / Clarifications / <i>Examples</i>
01 Processes / Procedures (Manual)	A series of actions, changes, or functions that when completed create a desired result: <i>The series of actions done to create a marketing campaign. A set of established forms or methods for conducting the affairs of a business.</i> <i>The system has to accurately calculate the Harmonized Sales Tax (HST) for all transactions originating in the Maritime provinces.</i>
11 Knowledge	Information and/or skills applied to a specific situation. This is the human (subjective) need that drives all IT applications. <i>How to change a tire.</i> <i>I want to know when to reorder training material from our vendor.</i>
12 Information	A collection of <u>related</u> facts or related data that is greater value than the individual datum. <i>Statistical information, customer profile.</i> <i>Our website ranks # 2 in worldwide access when searching for the term "Business Requirements Gathering".</i>
13 Data Descriptions	The properties or characteristics of an element of data. <i>The data element "customer" has a definition, character length, etc.</i> <i>A customer is an individual or organization that has purchased products or services from us in the past 2 years. The customer-id is a unique, 15-digit, positive, real number that identifies a single customer.</i>
14 Help facilities	Context specific information about how to perform a function, action or process of a system. <i>The user has to be able to see the corporate standard definition of every element on a data entry screen.</i> <i>The system date can be set to these formats: mm/dd/yyyy, dd-Month Name-yyyy.</i>
15 Data Persistence	The length of time that a data element (or group of data) must be kept available including the media of retention. <i>Legal, organizational, operational, or other needs defining the life expectancy of data.</i> <i>Results of a medical trial must be kept electronically available for fourteen years after the trial is completed.</i>
21 Urgency	The length of time a person can wait on a response from the automated system before their ability to do their job becomes impaired. <i>The desired time lapse for the next event to occur after a specific event occurs.</i> <i>Our customer service representatives have to know the credit status of a caller within 5 seconds of identifying the customer.</i>

Requirements Taxonomy

Business Requirements	Definitions / Clarifications / <i>Examples</i>
22 Frequency	The number of instances an event or process occurs in a specific interval of time. <i>Transactions/functions per unit of time</i> <i>Payroll will be processed every two weeks.</i>
23 Volume	The amount of data (information) stored, transmitted, presented, and/or used by a process. <i>Number of occurrences of records or items on a report</i> <i>We anticipate an active customer base of 10,000 by the end of 2006.</i>
24 Accuracy	The acceptable deviation from a defined standard or target. <i>Tolerance for errors from the business community</i> <i>The on-hand inventory file values must be within +/- 2% of the actual physical count.</i>
31 Usability	A measure of how easy (intuitive) it is to use the system. Often referred to as user-friendliness. <i>Subjective dimension that usually requires prototyping or visual representation to clarify</i> <i>A trained user should be able to complete an average of 35 reservations per day.</i> <i>The graphical interface will not depend upon color alone to convey mission-critical information.</i>
32 Trainability	A measure of how easy it is to learn how to use the system. <i>Desired length of training time to achieve a specific goal</i> <i>After three hours of web based training and one hour of practice, the user will be able to correctly process 8 out of 10 map requests in less than 2 hours.</i>
33 Flexibility	A measure of how easy it is to add new functionality or change existing functionality <i>Length of time expected to accomplish maintenance task</i> <i>A support programmer with 1 year of experience supporting this system should be able to create a new two page (screens) user view in less than two hours.</i>
34 Availability	A measure of the planned up time during which the system can be used. It is usually expressed as a percentage of duration. <i>Mean time between failures and accumulated down time during failures</i> <i>During peak time (10am to 2pm) the system be operational 99.5% of the time.</i> <i>The nuclear power plant radiation level monitor function has to be available 24/7/365.</i>

Requirements Taxonomy

Business Requirements	Definitions / Clarifications / <i>Examples</i>
41 Distribution (Organization)	The roles and responsibilities of the users of the system. <i>Who can do what or gets what information</i> <i>A clerk will be able to determine inventory from any of the floor access terminals.</i>
42 Location	The physical site of the components of the system (including people). <i>Geographic or structure of hardware, software, resources</i> <i>The application will be accessible from any city in Florida.</i>
43 Security	A description of the characteristics of those people who are allowed to use the system and the means by which they gain access to the system. <i>Use of secret codes and/or biometric identifiers</i> <i>Only E8 level and above personal can view the Andromeda files based on retinal scan and fingerprint id.</i>
44 Legal	Specific laws or regulations with which the system must comply. <i>Externally mandated constraints</i> <i>Personal health data will not be stored with personal identity data.</i>
45 Cultural	Human factors that must be consider in the design of the system. It usually denotes a personal quality resulting from the development of intellect, manners, and aesthetic appreciation. <i>Localization considerations</i> <i>The system cannot use any red fonts.</i> <i>Information on the order entry interface has to have the same meaning in English, German, French and Italian.</i>

Requirements Taxonomy

Technical Specifications	Definition / Example
51 Modules / Programs (Automated)	These technical specifications describe what each module or program will do. In the object world, this category specifies assemblies, classes, applications, etc. <i>HSTo1 will calculate HST for a single toll call and round it to the nearest cent.</i>
52 Architectural / Structural	Architectural and structural specifications indicate how the application will be built. <i>The Order Entry system will be MS windows-based.</i>
61 Data structures	Data structure specifications describe the groupings of the information that may be needed by people or by applications. A data structure can be a flat file, a table, an array, a property or set of properties (for a class), or any other collection of basic raw data that is structured in a manner to be usable by an application. <i>The CustOrder file contains CustID, CustName, OrdDate, OrdQty, and OrdPrice elements.</i>
62 Data bases	Database specifications determine the technology that will be used to maintain the groupings of data structures. A database is a structured group of tables, relationships, data values, dated a script since, and other information that describes its structure and contents. <i>The Cash Forecasting database will be developed in Oracle 8i.</i>
63 Access Paths	Access paths specifications describe primary keys and relationships that determine how data can be accessed by programs or modules. <i>The CustomerID, ZipCode, LastOrderDate will be used to access the most current credit ratings for a customer.</i>
64 User Views	User view specifications described collections of information structured and presented in a manner that is people friendly. Typically, there are two levels to user views, the data level and the presentation level. Both levels need to be defined independent of each other. <i>See the AddBook screen prototype.</i>
65 Data Properties	Data properties specifications are commonly referred to as metadata). They encompass any information that is needed to describe length, content type, constraints, ownership, or any other dimension of the information that is being manipulated. <i>CustType: Numeric - 1=Public, 2=Corporate, 3=Individual</i>
71 Size	Size specifications are quantitative measures that can refer to the size of the database (as in the number of all records of each type) or the size of an application (as in how many MB or GB the distributed application requires). <i>The recorded seminar presentation will not exceed the capacity of a CD-Rom (720 MB)</i>
72 Precision	Precision is specified as a measure of accuracy as it applies to informational content or the results of calculations. It is typically expressed in terms of a number of places behind the decimal point. <i>HST will be rounded to the nearest cent.</i>

Requirements Taxonomy

Technical Specifications	Definition / Example
73 Response Time	Response time specifications mandate how quickly the automated system has to respond to a user request. <i>Peak response time for credit rating will not exceed 1.5 seconds.</i>
74 Input Urgency	Input urgency specifies how quickly a component of the application needs input from elsewhere before it is incapable of completing its mission. Typically, sub-second input urgency implies the need for a real time application. In interactive applications the input urgency is often measured in minutes whereas in batch systems it is commonly ignored. <i>The reactor shut-off module will shut the system down if it does not receive an override within 3 seconds after issuing a temperature exceeded warning.</i>
81 Maintainability	Maintainability specifications express how easy it has to be to modify the behavior of an application once it has been deployed. Concepts such as cohesion and coupling are measures of maintainability. <i>It has to be possible to add new loan rate calculation modes to the system with less than 1 developer day of effort.</i>
82 Reliability	Reliability specifications are an expression of how errors that the application makes affect the outside world. Common measures of reliability are mean time between failures (MTBF), mean time to repair (MTTR), etc. <i>The patient monitoring system down time will not exceed 1 cadaver between failures.</i>
83 Portability	Portability specifications are necessary for applications that will be installed on multiple operating systems. This measure can be critical for applications that are used in n-tiered architectures and/or on the internet/intranets. <i>The application will function without modifications on Windows, Linux, Apache and Apple operating systems.</i>
84 Scalability	Scalability specifications are needed to define acceptable criteria for an application to grow over time in reaction to growing business events or volumes of data. <i>The system has to accommodate a 500% annual increase in sales.</i>
91 Interfaces (Interaction)	Interfaces/interaction specifications of the boundary between the users and automation or between applications that need to communicate with each other. <i>The payroll system has to communicate with the HR system using EDI format 8II6B.</i>
92 Standards	Standards specifications determine which corporate, industry-wide, best practice or other limits will be imposed on the structure, content or context of the evolving solution. <i>Date will be represented as dd-MonthName-yyyy (02-September-2066).</i>

Requirements Taxonomy

Technical Specifications	Definition / Example
93 Infrastructure	Infrastructure specifications describe the hardware/software configuration that is needed for the project to be successful. This can also include system software, middleware, and other necessities. <i>The disk drives of the system will be shadowed at each regional center and the global center.</i>
94 Methodology	Methodology specifications identify the approach that the project will follow and why. <i>THE Guide Project Management and System Development methodology will be used for this project.</i>
95 Technology	Technology constraints limit the project teams' ability to make decisions regarding hardware / software platforms for their particular application. <i>The application has to run on the existing hardware/software platform.</i>